

JomTravel by ZooNegara

Team: Jackson Yeoh, Tan Wei Ni

Problem Statement: Travel Planner

Video Presentation: [<https://youtu.be/LBJMipqd9Jc>]

Presentation Slides: [<https://canva.link/0n2v3z0hi5cc2s2>]

1. Project Overview

The Problem.

Planning a trip involves coordinating flights, accommodation, transportation, activities, budgets, and personal preferences. Travellers often have to switch between multiple platforms to compare prices, make bookings, organise itineraries, and communicate with their travel companions. This becomes even more challenging for group trips, where different people have different budgets, interests, schedules, and expectations.

The main causes of this problem include:

- Fragmented travel planning tools that handle only one part of the travel experience.
- Difficulty comparing affordable flights and accommodation while keeping track of total spending.
- Time-consuming itinerary planning that does not always account for travel distance, opening hours, or personal interests.
- Poor coordination between group members when deciding activities and sharing expenses.
- Limited support when unexpected events, such as flight delays, bad weather, or cancelled activities, disrupt the original itinerary.
- Lack of engagement and motivation in the planning process, which can make organising a trip feel like a chore.

Stakeholders

Stakeholder	Needs
Solo travellers	Affordable travel options, personalised itineraries, and convenient trip management.
Group travellers	Shared planning, preference matching, communication, and expense splitting.
Budget-conscious travellers	Flight and accommodation comparisons with clear cost estimates.
Travel service providers	Opportunities to direct users to booking platforms.
Friends and travel companions	An easy way to coordinate activities and share experiences.

Existing market solutions and their limitations

Existing app	What it offers	Why it falls short
Skyscanner	Flight and travel price comparison.	Primarily focuses on searching travel deals rather than providing a complete shared travel experience.
Booking.com	Accommodation and travel bookings.	Strong booking functionality, but users may still need separate tools for group coordination, quests, and a unified trip workspace.
Google Maps	Maps, places, navigation, and saved locations.	Useful for discovering places and navigation, but not designed as a complete budget-first group travel planner with game mechanics.
Wanderlog	Trip itineraries, maps, and collaborative travel planning.	Provides many planning features, but our concept focuses on combining deal comparison, group preference matching, and a gamified travel experience.

The gap: Travellers need to manually combine information from different platforms. Our project aims to connect the planning, booking references, budgeting, group coordination, and actual travel experience into one platform.

Our Solution.

TripQuest is an integrated travel planning platform designed to make organising a trip faster, more affordable, and more enjoyable. Users can compare flight and accommodation options, set a travel budget, and generate personalized itineraries based on their interests, schedules, and destinations. For group trips, users can invite friends to collaborate, vote on activities, share expenses, and communicate within a shared trip room. During the journey, TripQuest provides a

central dashboard for bookings, maps, checklists, and travel quests, while AI-assisted re-planning helps users adapt when unexpected changes occur.

2. Ideation & Process

2.1 Ideas We Considered

The following table organises our initial brainstorming ideas. The selected features are listed first, followed by ideas that were considered but narrowed or dropped for the prototype.

Idea	Why it was dropped / kept
Flight & accommodation comparison (Chosen)	Kept because affordable travel options directly support the budgeting problem and provide a strong starting point for the platform.
AI itinerary generator (Chosen)	Kept because it helps users plan activities based on interests, budget, dates, and location, reducing planning time.
Shared trip dashboard (Chosen)	Kept because it connects bookings, maps, budgets, and itineraries into one accessible workspace.
Group collaboration & preference matching (Chosen)	Kept because coordinating different preferences and budgets is a major challenge in group travel.
Travel quests / gamification (Chosen)	Kept because it differentiates our product from conventional travel planners and encourages engagement during the trip.
Pre-trip checklist & reminders (Chosen)	Kept because preparation is an important part of travelling and helps users avoid forgetting essential tasks.
AI re-planning for unexpected changes (Chosen)	Kept as a key differentiator, with a focused prototype scenario such as a delayed flight or rainy day.
Built-in group chat (Chosen — scoped)	Kept as a lightweight collaboration feature. A full-featured messaging system is outside the initial build scope.
Real-time flight and hotel booking APIs	Narrowed for the prototype. Live pricing and availability are useful, but API access, rate limits, and booking integration can increase complexity.
Full direct booking system	Dropped from the initial build. The prototype will use comparison results and booking links instead of handling payments and reservations directly.
Full social media travel platform	Dropped because it expands the product beyond the core travel planning problem.
Advanced AR navigation	Dropped because it requires additional development

	and is not essential to demonstrate the solution.
AI-generated travel videos	Dropped because it does not directly solve the main planning and coordination problems.
Complex location-verified quest system	Narrowed to simple check-ins or photo submissions for the prototype.

2.2 Ideation Boards

Ideation Board 1 — Problem Tree / Affinity Diagram

Problem Tree

This board shows how the team identified the causes of stressful travel planning, including fragmented apps, budget uncertainty, group disagreements, and unexpected changes. The ideas were grouped into budgeting, planning, collaboration, and travel experience.

Ideation Board 2 — Feature Mindmap

Feature Mindmap

This board organises the possible features of TripQuest, showing how flight comparison, AI itineraries, shared trips, checklists, travel quests, and re-planning connect to the overall travel experience.

Ideation Board 3 — User Flow

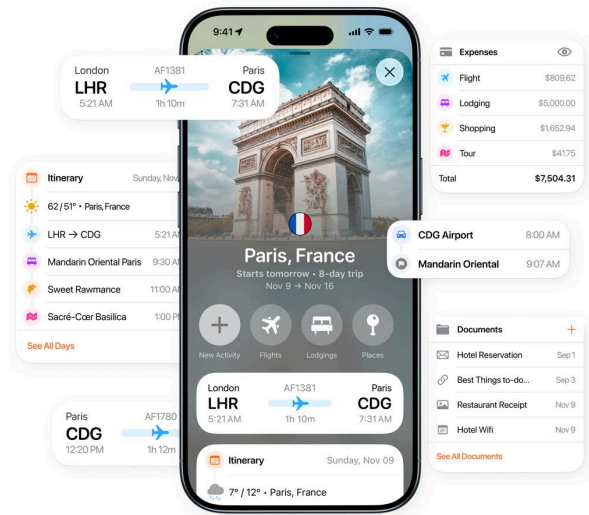
User Flow

This board illustrates the intended user journey, from creating a trip and comparing prices to collaborating with friends, travelling, completing quests, and adjusting plans.

3. Design & Prototype

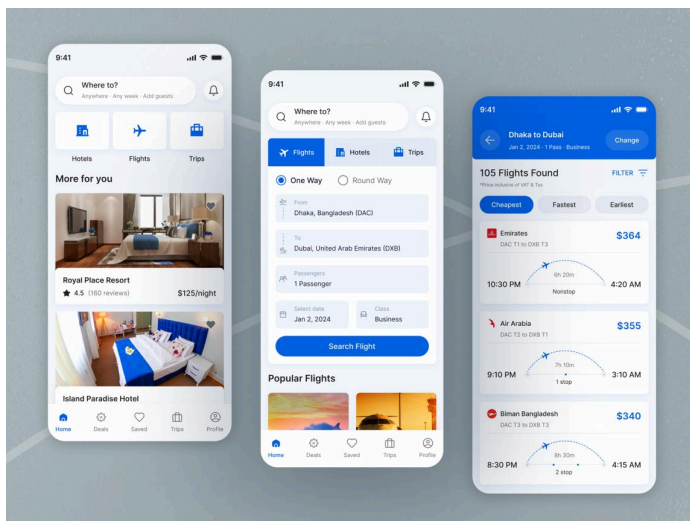
The TripQuest prototype will demonstrate the main user journey from planning a trip to managing it during travel. The design will prioritise a simple and intuitive interface that allows users to access important trip information without switching between multiple applications.

Key Screens



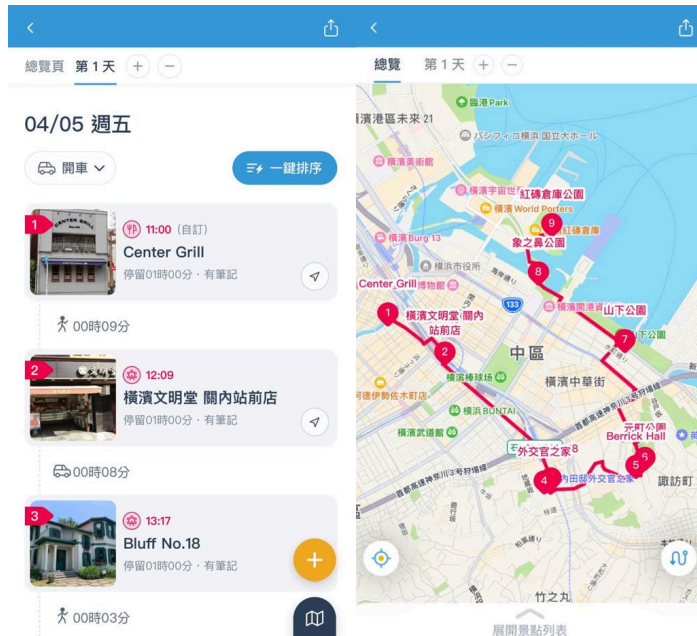
Screen 1 — Trip Dashboard

The main overview of the trip, displaying travel dates, total budget, upcoming activities, saved bookings, and overall progress.



Screen 2 — Flight & Accommodation Comparison

Users can compare options based on price and preferences, select a booking, and add its cost to their trip budget.

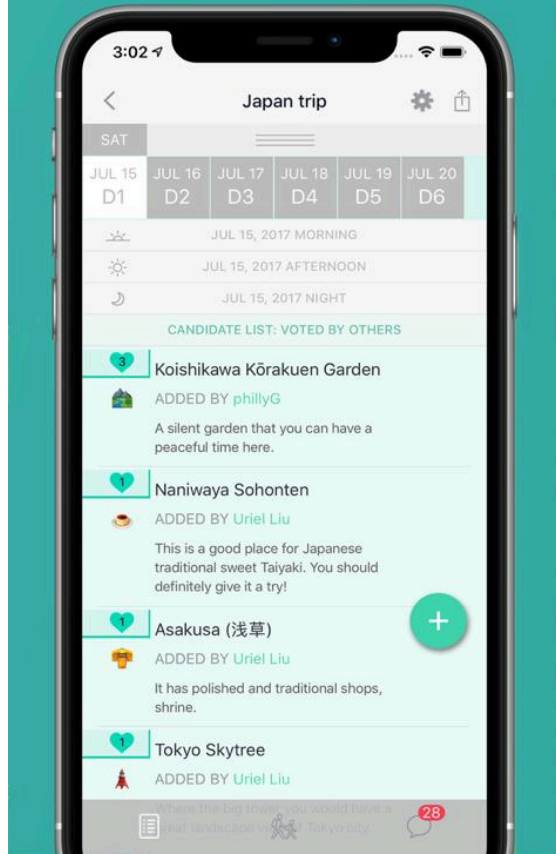


Screen 3 — AI Itinerary Builder

Users enter their destination, budget, travel dates, and interests to generate a structured itinerary with locations and activity details.

Collaborate without chaos

Get a quick consensus from poll,
as you build the itinerary.



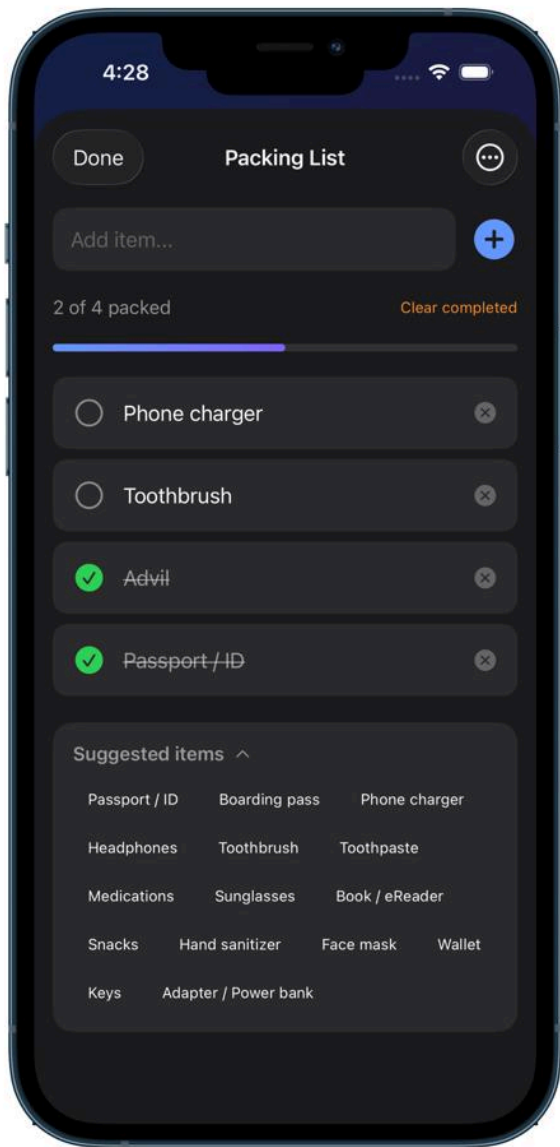
Screen 4 — Group Trip Room

Friends can join the trip, view shared plans, vote on activities, and communicate within the group.



Screen 5 — Travel Quest & Check-in

Users can complete activities, collect points, and track their progress throughout the trip.



Screen 6 — Pre-Trip Checklist

A checklist for booking confirmation, documents, packing, and other preparation tasks, with reminder functionality.

4. What Makes It Different

TripQuest combines travel planning with an interactive group experience. Instead of functioning as a standalone booking search tool or itinerary generator, the platform aims to support the entire travel journey.

Feature	What makes it different
Budget-connected travel search	Flight and accommodation selections are connected to the trip budget, allowing users to see how each decision affects total spending.
Booking-to-itinerary workflow	Saved booking and map links can be connected to a daily itinerary, making the information useful both before and during travel.
Group preference matching	The platform brings together individual budgets and interests to help groups make decisions collaboratively.
Travel quests & gamification	Users can turn real travel activities into challenges, collect points, and create shared memories with friends.
AI-assisted re-planning	The itinerary is designed to be adaptable when unexpected events affect the original schedule.
One central travel workspace	Users can access their bookings, maps, budgets, checklists, group plans, and quests from one trip dashboard.

Our key differentiator

TripQuest transforms travel planning from a stressful coordination task into a shared adventure. The combination of budget-conscious planning, group collaboration, and travel quests is the central twist of our solution.

5. Technical Architecture & Feasibility

Tech stack

The following is a proposed stack for the prototype. The team can adjust it based on available skills and competition requirements.

Technology	Purpose & Reason
React + Vite	Frontend development. Allows the team to build reusable UI components and an interactive web application efficiently.
Tailwind CSS	Styling and responsive design. Helps the team create a consistent interface quickly.
Supabase	Backend, PostgreSQL database, authentication, and real-time features. Useful for managing trips, members, itineraries, and shared data.
AI API	Generates itineraries, suggests activities, and supports itinerary

	re-planning.
Maps API	Displays locations, map links, and travel destinations.
Flight / accommodation API	Provides live or sample travel comparison data, depending on API access.
Vercel	Frontend hosting and deployment. Suitable for hosting a web prototype.
GitHub	Version control and collaboration between team members.

Expected constraints

Live travel data

Real-time flight and accommodation data may require third-party API access, authentication, rate limits, and potentially paid services. For the competition prototype, the team can use mock data or a limited API integration to demonstrate the booking comparison workflow.

Direct booking

Handling actual reservations, payments, cancellations, and refunds would require substantial integration with booking providers. The initial prototype will redirect users to booking links rather than processing payments.

AI reliability

AI-generated itineraries may contain inaccurate prices, opening hours, or travel durations. The prototype should clearly label estimated information and allow users to edit or approve suggested plans.

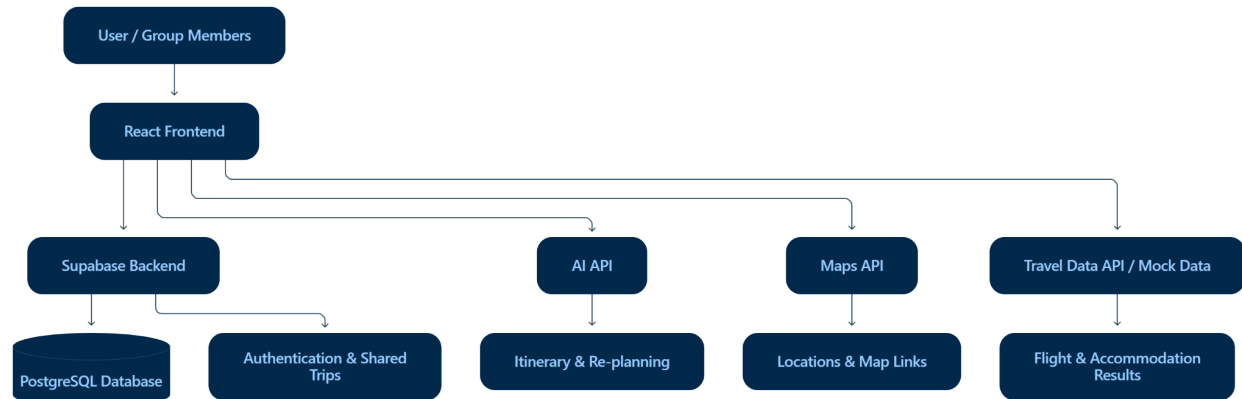
Real-time group chat

A full messaging platform requires additional backend functionality. For the initial scope, the team can implement shared comments or a lightweight chat room.

Maps and location permissions

Advanced check-in verification may require location permissions and additional testing. A simple manual check-in or photo-based completion system is more feasible for the prototype.

System architecture diagram



Build plan & scope

The building phase will focus on delivering a functional web prototype that demonstrates the complete travel planning workflow without attempting to build a full commercial booking platform.

Core functionality to build

Prototype scope

1. Create a trip with destination, dates, and budget.
2. Compare sample flight and accommodation options.
3. Save selected options and calculate estimated trip costs.
4. Generate a personalised itinerary based on interests.
5. Display itinerary activities with map links.
6. Create a shared trip room and invite group members.
7. Implement a simple activity voting or preference matching feature.
8. Add pre-trip checklist functionality.
9. Add a simple travel quest and check-in mechanism.
10. Demonstrate one AI re-planning scenario, such as a delayed flight.

Features outside the initial scope

- Full real-world flight and hotel booking transactions.
- Payment processing and automatic refunds.
- Complete social media functionality.
- Advanced AR navigation.
- Comprehensive real-time location tracking.
- Fully automated rebooking with external providers.

Proposed development phases

Phase	Tasks
Phase 1 — Ideation & UI	Finalise user flow, create wireframes, and design key screens.
Phase 2 — Frontend	Build trip creation, dashboard, comparison, and itinerary screens.
Phase 3 — Backend	Set up database, shared trips, and basic authentication.
Phase 4 — AI & APIs	Connect itinerary generation, mock travel data, and map links.
Phase 5 — Group & Game	Implement preference matching, checklist, and travel quests.
Phase 6 — Testing & Presentation	Test the main user journey, fix bugs, and prepare the demo and presentation.

Final prototype demonstration

The team will demonstrate a group of friends planning a 5-day Osaka trip:

1. Jackson creates a trip with a budget and travel dates.
2. The platform displays flight and accommodation options.
3. The selected travel options are added to the budget.
4. AI generates an itinerary based on the group's interests.
5. Friends join the trip room and vote on activities.
6. The group views the shared itinerary, map links, and checklist.
7. During the trip, members complete a travel quest.
8. A flight delay scenario triggers AI-assisted itinerary re-planning.

Success criteria: A user should be able to understand and complete the main travel planning workflow through the prototype, while the judges can clearly see how the solution reduces planning stress, improves group coordination, and makes travel more engaging.